Raed Mohamed Amine Hamrouni

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raed-med-amin-hamrouni | hamrouniraed07 | Portfolio

Profile

AI Software Engineer holding an Engineering degree in Data Science & Artificial Intelligence and a Master's degree in Web Services & Multimedia. Experienced in full-stack development, LLM integration, and microservices architecture. Proficient in Python, Node.js, React, FastAPI, Docker, and AI models (Gemini, Llama, Claude). Passionate about building intelligent, production-ready systems that solve real-world problems.

Education

Polytechnic School of Sousse, Engineering Degree **2024 – 2026**
- Specialty: Data Science & Artificial Intelligence

ISITCOM – Sousse, Professional Master's Degree **2023 – 2026**
- Specialty: Web Services and Multimedia

ISITCOM – Sousse, Bachelor's Degree in Computer Science **2020 – 2023**
- Specialty: Computer Science and Multimedia

Professional Experience

AI Software Engineer Intern (Final Engineering Cycle Project), ClassQuiz – Sousse, Tunisia **Jan 2026 – Jun 2026**
- Built ClassQuiz, a production-ready AI-powered exam correction platform for primary schools, covering the full pipeline from admin upload to automated PDF report generation.
- Integrated **Gemini 2.5 Flash** for Arabic handwriting OCR and **Llama 3.2** (Ollama, local) for automated grading and pedagogical feedback with a confidence-based validation pipeline (threshold: 70%).
- Technologies: Node.js, React, Python, FastAPI, Gemini 2.5 Flash, Llama 3.2, MongoDB, Docker, JWT

Data Science Intern, ARSII – Sousse, Tunisia **Jul 2025 – Sept 2025**
- Designed a supervised classification model to predict postoperative complications following pediatric circumcision, enabling early identification of risk factors and improved medical follow-up.
- Technologies: Python, TensorFlow, NumPy, Pandas

Software Developer Intern (Bachelor's Final Project), VIVIA – Sousse, Tunisia **Jan 2023 – Jun 2023**
- Developed a customer complaints management module for the company's CRM system, reducing complaint processing time significantly.
- Technologies: Node.js, React.js, PostgreSQL, Express.js

Projects

RAG System (Medical Domain) – GitHub **Jan 2026 – Apr 2026**
- Designed and implemented a Retrieval-Augmented Generation (RAG) system for querying medical documents with source-based answers.
- Built a fully local AI pipeline using FAISS for vector search, HuggingFace embeddings, and Ollama (LLM) for inference.
- Developed CLI interface, Streamlit web application, and evaluation scripts for retrieval precision and answer accuracy.
- Technologies: Python, LangChain, FAISS, HuggingFace, Ollama, Streamlit

Credit Card Fraud Detection – MLOps – GitHub **Nov 2025 – JAN 2026**

- End-to-end MLOps pipeline for fraud detection on 284K transactions — 99.9% precision, 97.8% AUC-ROC, <10ms API response time.
- Automated training (ZenML + MLflow), hyperparameter tuning (Optuna), REST API (FastAPI), monitoring (Prometheus + Grafana) and CI/CD (GitHub Actions).
- Technologies: Python, FastAPI, ZenML, MLflow, Optuna, Docker, Prometheus

FitBox – AI Fitness Coach – GitHub **Oct 2025 – Jan 2026**

- Intelligent fitness and nutrition coaching application powered by AI, providing personalized workout plans, nutritional advice, and conversational coaching using Llama 3.2 via Ollama with LoRA fine-tuning.
- Technologies: Python, Flask, Streamlit, Ollama, Hugging Face Transformers, Pandas, PyTest

FoodOrder – AI-Powered Food Delivery Platform (MERN Stack) – GitHub **Sep 2025 – Dec 2025**

- Developed a full-stack food delivery platform inspired by UberEats using the **MERN stack** (MongoDB, Express.js, React, Node.js), supporting restaurant browsing, meal ordering, cart management, and order tracking.
- Integrated an **AI recommendation system** using **Ollama (Llama 3.2)** to provide personalized restaurant suggestions, meal recommendations, and multi-criteria comparisons based on user preferences.
- *Tech: MongoDB, Express.js, React, Node.js, Redux Toolkit, JWT, Ollama (Llama 3.2), Axios*

Biological Age Estimation – GitHub **Feb 2025 – May 2025**

- Regression model to predict biological age from dermatological data.
- Technologies: Python, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, Matplotlib, Seaborn

Technical Skills

Languages: Python, JavaScript, TypeScript, R

Frontend: React.js, Vite, TanStack Query, Zustand, Redux, Tailwind CSS

Backend: Node.js, Express.js, FastAPI

Databases: PostgreSQL, MySQL, MongoDB, Mongoose

AI & LLM: Gemini, Ollama, OpenAI, Hugging Face, LoRA fine-tuning, Prompt Engineering, RAG

Machine Learning: Scikit-learn, TensorFlow, Keras, PyTorch, ZenML, MLflow, Optuna

DevOps & Architecture: Docker, Kubernetes, REST, GraphQL, gRPC, Apache Kafka, JWT

Data Science: Pandas, NumPy, Matplotlib, Seaborn, supervised/unsupervised learning, feature engineering

Tools: Git/GitHub, VS Code, Postman, Jupyter Notebook, Google Colab

Certificates

- **Artificial Intelligence Fundamentals** — IBM **2025**
- **Introduction to R** — DataCamp **2025**
- **Intermediate R** — DataCamp **2025**
- **Introduction to Tidyverse** — DataCamp **2025**
- **Cleaning Data with Generative AI** —DataCamp **2025**
- **Introduction to AI Agents** — DataCamp **2025**
- **AI-Assisted Travel Planning** — DataCamp **2025**

Languages

Arabic: Native **French:** Advanced **English:** Advanced **German:** Beginner